

# CE 310

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## Probability and Bayes' Theorem

Week 3 · Classical Probability · Conditional Probability · Bayes

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# Part 1

Why Conditioning Changes Everything

# The Launch That Should Not Have Happened

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January 28, 1986 · Space Shuttle Challenger · 28°F at launch

- O-ring seals failed in the cold → shuttle broke apart 73 seconds after launch
- Seven crew members died
- Engineers had raised concerns the night before



*73 seconds after launch. NASA/KSC — public domain.*

**What went wrong:** failure data was presented as **overall O-ring damage rates**, not conditioned on temperature. Plotted against launch temperature, the pattern was unmistakable.

⚠ P(O-ring damage |  $T < 53^{\circ}\text{F}$ ) was far higher than P(O-ring damage) overall. The marginal probability obscured the conditional risk. NASA launched anyway.

# Three CE/ArcE Decisions — All Require Conditioning

| Design Decision                 | Wrong approach                                | What you actually need                                                         |
|---------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------|
| Size HVAC cooling system        | Annual average cooling (19 kWh/hr)            | $P(\text{Demand} > \text{threshold} \mid \text{summer business hours})$        |
| Size backup generator           | Marginal $P(\text{Electricity} > 100) = 39\%$ | $P(\text{Electricity} > 100 \mid \text{business hours}) \approx \mathbf{96\%}$ |
| Schedule preventive maintenance | Overall fault rate                            | $P(\text{Fault} \mid \text{sensor reading anomalous})$                         |

💡 The third row is different: you know what the sensor says, and you want to know what that means about the equipment. That requires **reversing** the conditional — computing  $P(\text{fault given alert})$ , not  $P(\text{alert given fault})$ . That is Bayes' theorem.

## Where We Left Off — A Marginal Probability

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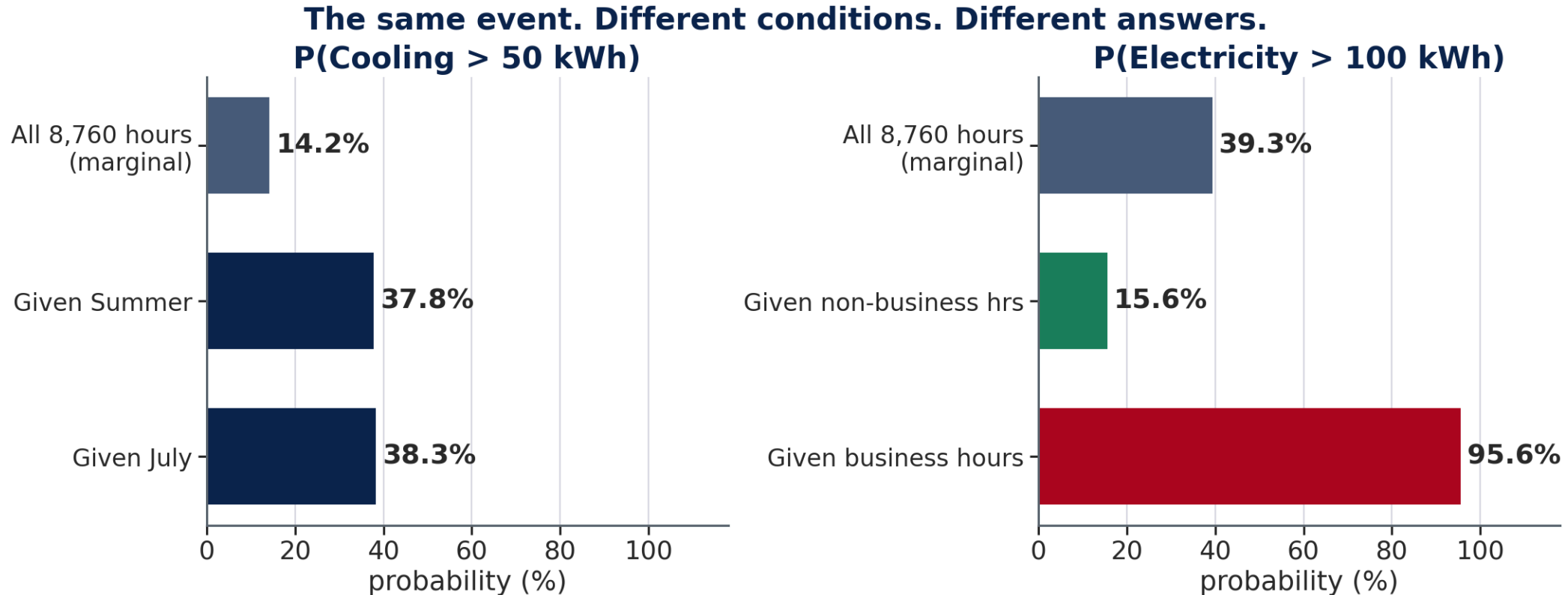
In the Tucson 2023 dataset, you counted 900 hours where  $T > 95^\circ\text{F}$ . Using that count:

$$P(T > 95^\circ F) = \frac{900 \text{ hours}}{8,760 \text{ hours}} = \mathbf{10.3\%}$$

This is a **marginal probability** — no conditioning on month, time of day, or occupancy.

**Week 3 takes the next step:** instead of  $P(\text{event})$ , we ask  $P(\text{event} \textbf{ given } \text{some context})$ . And then we ask what happens when you need to flip that conditional around.

# Conditioning Changes the Answer — Not a Little



# AI / ML Bridge — Conditional Frequencies

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A building fault detector computes:

$$P(\text{fault} \mid \text{cooling spike}) = \frac{P(\text{cooling spike} \mid \text{fault}) \cdot P(\text{fault})}{P(\text{cooling spike})}$$

| Term                                        | What it means                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------|
| $P(\text{fault} \mid \text{cooling spike})$ | <b>Target:</b> given the alarm triggered, is there a real fault?            |
| $P(\text{cooling spike} \mid \text{fault})$ | Sensor reliability — how often does the alarm fire when a fault is present? |
| $P(\text{fault})$                           | Baseline fault rate — how common are faults in normal operation?            |
| $P(\text{cooling spike})$                   | Overall spike rate — how common are spikes with or without a fault?         |

This is Bayes' theorem applied to the conditional frequency tables you build today with COUNTIFS.

# Part 2

## Classical Probability



# What Is Probability?

 **Probability:** A number between 0 and 1 that quantifies how likely an event is to occur.  $P = 0$  means impossible;  $P = 1$  means certain.

## Three interpretations — all valid in engineering:

### Frequency

$P(A)$  = fraction of times  $A$  occurs over many repeated trials

*"This chiller faults in 5% of operating hours"*

### Classical

$P(A)$  = favorable outcomes / total equally likely outcomes

*" $P(\text{Monday}) = 1/7$  for a randomly chosen weekday"*

### Subjective

$P(A)$  = degree of belief, updated with evidence

*"I believe there is a 70% chance this component needs replacement"*

# Sample Space, Events, and the Three Rules

**Sample space  $\Omega$ :** the set of all possible outcomes

| Rule                      | Formula                                               | Plain meaning                                              |
|---------------------------|-------------------------------------------------------|------------------------------------------------------------|
| <b>Complement</b>         | $P(A^c) = 1 - P(A)$                                   | If $P(\text{rain}) = 0.3$ , then $P(\text{no rain}) = 0.7$ |
| <b>Addition</b>           | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$             | Probability of A <b>or</b> B (or both)                     |
| <b>Mutually exclusive</b> | $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$ | A and B cannot both occur                                  |

 **Key distinction:** The addition rule subtracts the overlap  $P(A \cap B)$  to avoid double-counting. When events are mutually exclusive the overlap is zero — so the simpler formula applies.

## Your Turn — Addition Rule

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A building has two independently monitored systems:

- $P(\text{HVAC fault in any given week}) = \mathbf{0.05}$
- $P(\text{Electrical fault in any given week}) = \mathbf{0.03}$

**What is  $P(\text{at least one fault occurs this week})$ ?**

*Think before calculating. A colleague says "just add them —  $5\% + 3\% = 8\%$ ." Is that right?*

## ✓ Addition Rule — Answer

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$$\begin{aligned} P(\text{some fault}) &= P(\text{HVAC}) + P(\text{Elec}) - P(\text{HVAC} \cap \text{Elec}) \\ &= 0.05 + 0.03 - (0.05 \times 0.03) = 0.08 - 0.0015 = \mathbf{0.0785} \end{aligned}$$

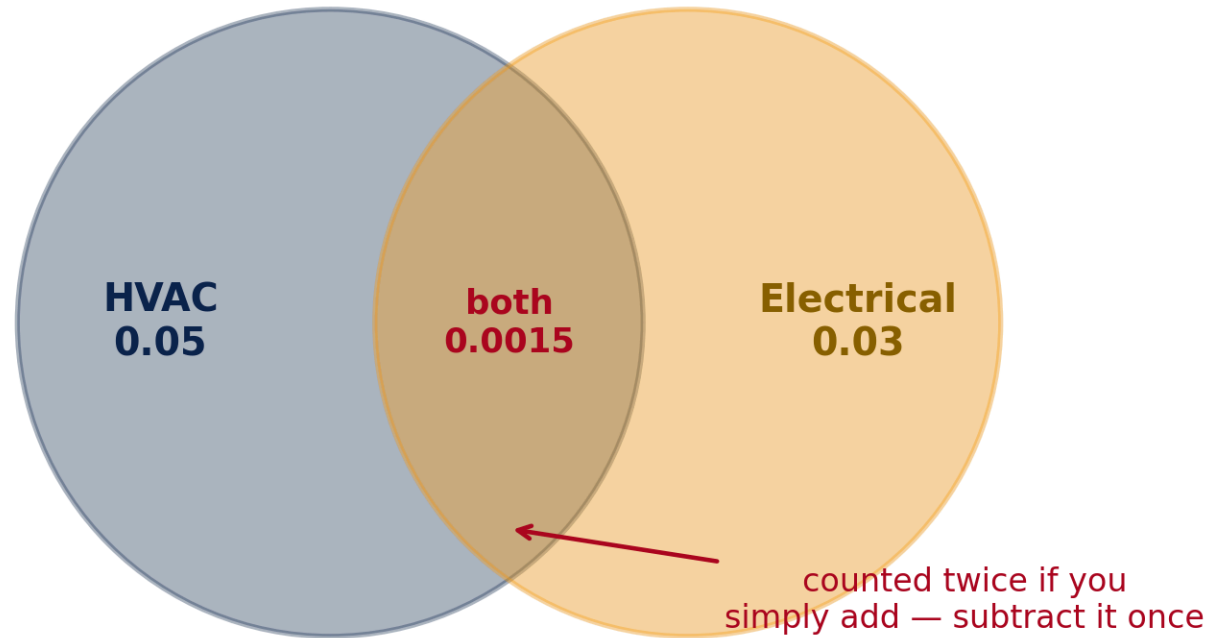
**Not 0.08** — the overlap (both systems failing in the same week) is double-counted if you simply add, and must be subtracted.

💡 The gap is small here (0.0015). In reliability systems with many components, uncorrected overlaps compound — and the error grows large enough to matter for safety calculations.

# Why You Subtract the Overlap

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$$P(A \cup B) = 0.05 + 0.03 - 0.0015 = 0.0785, \text{ not } 0.08$$



# Quick Check — Which Rule Applies?

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## 90 seconds · three questions

A building automation system monitors two subsystems independently:

- $P(\text{HVAC fault in any given week}) = 0.10$
- $P(\text{Lighting fault in any given week}) = 0.04$

**Question A:** What is  $P(\text{no HVAC fault this week})$ ?

**Question B:** What is  $P(\text{HVAC fault OR lighting fault this week})$ ?

**Question C:** A colleague says "just add them —  $10\% + 4\% = 14\%$ ." When is that answer exactly right, and when is it wrong?

## ✓ Quick Check — Answer

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**A:**  $P(\text{no HVAC fault}) = 1 - 0.10 = \mathbf{0.90}$

**B:**  $P(\text{HVAC OR lighting fault}) = 0.10 + 0.04 - (0.10 \times 0.04) = \mathbf{0.136}$

**C:** Adding directly gives the right answer **only when the events are mutually exclusive** — when they cannot both happen at once. Since both faults can occur in the same week, there is an overlap (both fault simultaneously with probability  $0.10 \times 0.04 = 0.004$ ) that must be subtracted.

💡 **Mutually exclusive vs. independent:** Mutually exclusive means they cannot both occur —  $P(A \cap B) = 0$ . Independent means knowing one happened tells you nothing about the other —  $P(A \cap B) = P(A) \times P(B)$ .

# Part 3

## Conditional Probability



# What Does "Given" Mean?

 **Conditional probability:**  $P(A \mid B)$  is the probability of A given that B has already occurred — or given that we restrict our view to the subset of outcomes where B is true.

**Formula:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{Count}(A \text{ AND } B)}{\text{Count}(B)}$$

**Intuition:** You started with the full sample space  $\Omega$ . Conditioning on B shrinks your sample space to only the outcomes where B occurred. You then ask: within that smaller world, how often does A happen?

**Key implication:**  $P(A \mid B) \neq P(A)$  whenever A and B are **associated** — when knowing B changes what you expect about A.

# "Given" Means: Throw the Other Rows Away

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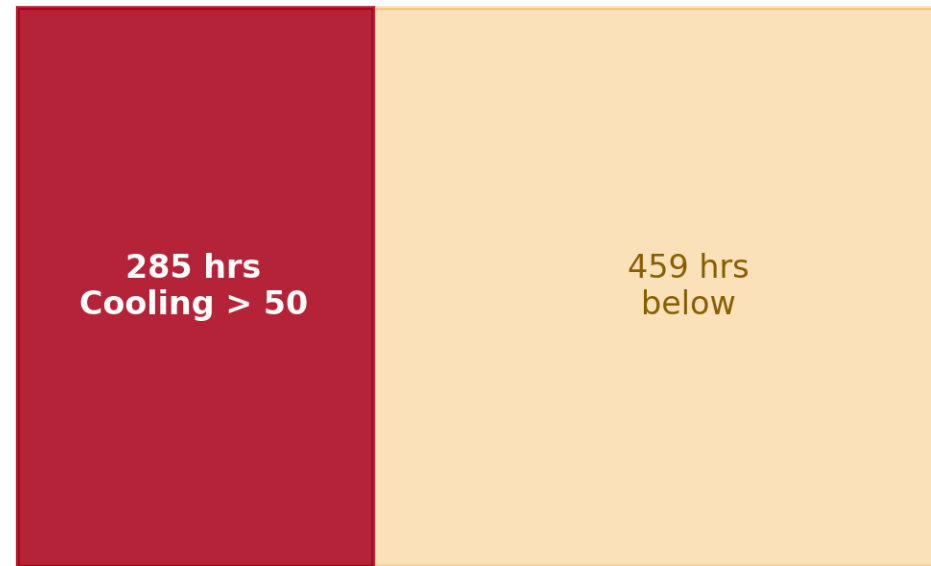
Conditioning shrinks the denominator — that is the whole idea

Step 1 — all 8,760 hours



July • 744 hrs

Step 2 — July becomes the whole world



$$P(\text{Cooling} > 50 \mid \text{July}) = 285 / 744 = 38.3\%$$

## Predict — Conditioning on Month

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In today's lab you will compute  $P(\text{Cooling} > 50 \text{ kWh} \mid \text{Month})$  for each month of 2023.

**The formula:**

$$P(\text{Cooling} > 50 \mid \text{July}) = \frac{\text{COUNTIFS}(\text{Cooling} > 50, \text{Month} = 7)}{\text{COUNTIF}(\text{Month} = 7)}$$

**What this does:** restricts the denominator to July hours only (744 hours), then asks what fraction of those July hours had high cooling demand.

**Overall probability (all 8,760 hours):**  $P(\text{Cooling} > 50 \text{ kWh}) = 14.2\%$

*Your prediction: what is  $P(\text{Cooling} > 50 \text{ kWh} \mid \text{Month} = \text{July})$ ? Higher? Lower? By how much?*

✓  $P(\text{Cooling} > 50 \text{ kWh} \mid \text{July}) = 38.31\%$

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**Marginal (all 8,760 hours):**

$$P(\text{Cooling} > 50 \text{ kWh}) = 14.2\%$$

**Conditional (July only):**

$$P(\text{Cooling} > 50 \text{ kWh} \mid \text{July}) = 38.31\%$$

Conditioning multiplies the probability by **2.7×**  
— nearly three times higher in July alone.

💡 The 14.2% is the right answer to the wrong question. For peak-load HVAC sizing, the right question is July (or July during business hours) — not the annual average. Conditional probability makes that gap visible.

# A Second Domain — Storm Drainage Design

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Conditioning is not an HVAC-only idea. A drainage engineer reviews 10 years of Tucson monsoon storm records (142 total storm events) while sizing a culvert.

**Marginal probability (any storm):**

$$P(\text{Culvert overtops}) = \frac{9}{142} = \mathbf{6.3\%}$$

**Conditioning on storm intensity:** of the 142 storms, 11 were classified as  $\geq 25$ -year design intensity. The culvert overtopped in 7 of those 11.

$$P(\text{Culvert overtops} \mid \text{storm} \geq 25\text{-yr}) = \frac{7}{11} = \mathbf{63.6\%}$$

💡 A culvert that "only overtops 6% of the time" overtops **10× more often** during the storms that actually stress it. Sizing to the marginal rate means the culvert fails nearly two-thirds of the time it matters most — the same mistake as the HVAC and generator examples, in a different discipline.

## Quick Check — Conditioning in a New Domain

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### 90 seconds

A transportation department reviews 3 years of crash reports at a signalized intersection (1,200 total observed days).

- Crashes occurred on 54 of the 1,200 days  $\rightarrow$  marginal  $P(\text{crash}) = 4.5\%$
- 210 of the 1,200 days had measurable rainfall
- Of those 210 rainy days, 27 had a crash

**Question:** Compute  $P(\text{crash} \mid \text{rain})$ . Is the marginal 4.5% a reasonable number to use when deciding whether to post a wet-weather speed advisory?

## ✓ Quick Check — Answer

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$$P(\text{crash} \mid \text{rain}) = \frac{27}{210} = \mathbf{12.9\%}$$

Compare to the marginal:  $P(\text{crash}) = 4.5\%$ . Conditioning on rain nearly **triples** the crash probability.

⚠ A wet-weather signage or speed-advisory program justified using the 4.5% marginal rate would badly understate the actual risk on the days it's meant to address. The same conditioning principle that governs HVAC sizing and generator sizing governs traffic-safety countermeasures.

# Predict Before You Compute

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The lab uses this conditioning formula:

$$P(\text{Electricity} > 100 \text{ kWh} \mid \text{BusinessHours} = \text{Yes})$$

Marginal  $P(\text{Electricity} > 100) = \mathbf{39.3\%}$

**Your prediction:** What do you think  $P(\text{Electricity} > 100 \mid \text{BusinessHours} = \text{Yes})$  is?

- **A)** About 39% — business hours probably doesn't matter much
- **B)** Around 55–65%
- **C)** Around 75–85%
- **D)** Above 90%



## ✓ The Answer — $P(\text{Electricity} > 100 \mid \text{BusinessHours}) \approx 96\%$

The marginal 39.3% is a **weighted average** of two very different conditional probabilities:

| Condition                        | $P(\text{Electricity} > 100 \text{ kWh})$ | Hours     |
|----------------------------------|-------------------------------------------|-----------|
| Business hours (Mon–Fri 8am–6pm) | $\approx 96\%$                            | 2,600 hrs |
| Non-business hours               | $\approx 16\%$                            | 6,160 hrs |
| <b>All hours (marginal)</b>      | <b>39.3%</b>                              | 8,760 hrs |

⚠ Sizing a backup generator for the 39.3% marginal probability means it will fail to cover business-hours demand 60% of the time — exactly when the building is occupied and the stakes are highest.

## Predict — Independence

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**Definition:** A and B are **statistically independent** if  $P(A \mid B) = P(A)$  — knowing B happened tells you nothing new about A.

**Equivalent test:** If independent:  $P(A \cap B) = P(A) \times P(B)$

**Scenario:** Two stormwater detention basins (A and B) serve adjacent sub-watersheds in the same city. Based on 10 years of storm records:

- $P(\text{Basin A overtops}) = 5\%$
- $P(\text{Basin B overtops}) = 5\%$

**What is  $P(\text{both basins overtop simultaneously})$ , if they are truly independent?**

## ✓ Independence in Practice — When It Fails

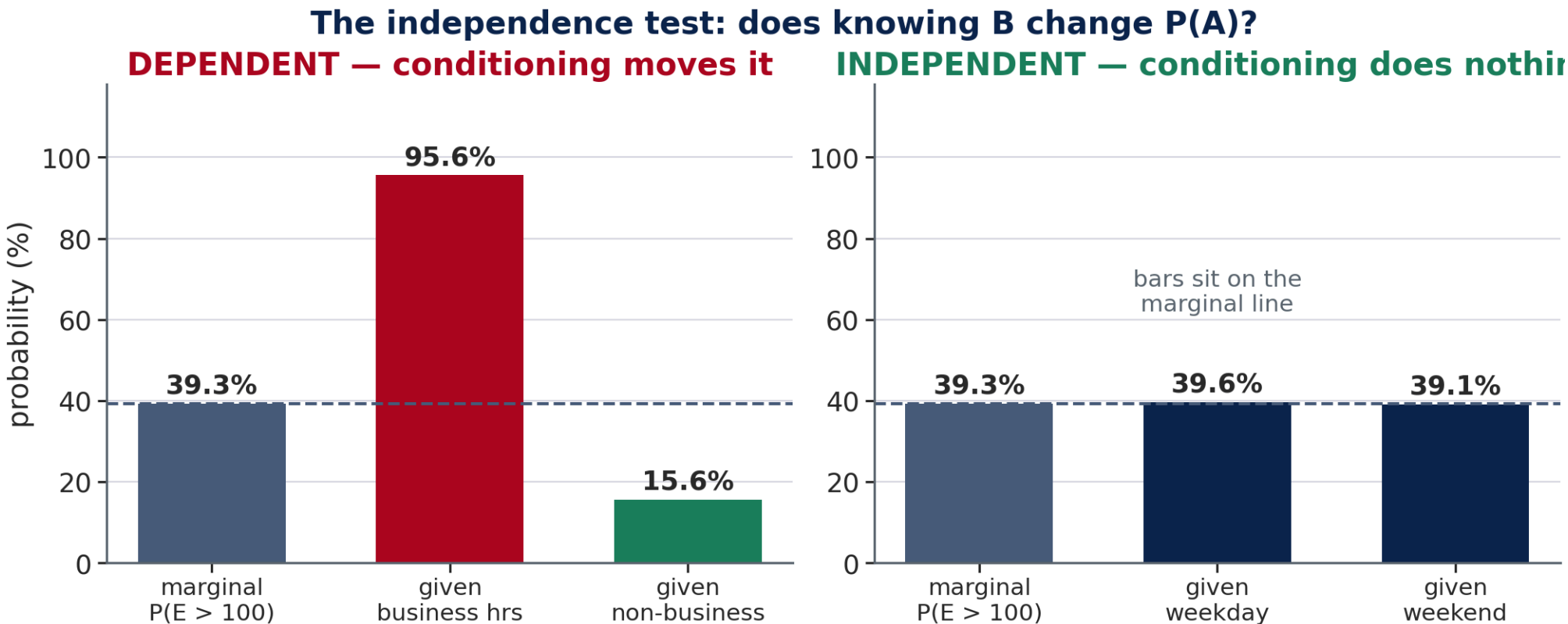
Observed over 10 years of storm events:

|                                    | Value        |
|------------------------------------|--------------|
| P(Basin A overtops)                | 5%           |
| P(Basin B overtops)                | 5%           |
| If independent: $0.05 \times 0.05$ | <b>0.25%</b> |
| <b>Actual observed</b>             | <b>4.1%</b>  |

**Actual rate is 16× higher** than independence predicts.

⚠ Both basins receive the **same storm**. Physically separate systems that share a common driver — the same storm, the same power grid, the same seismic event — are **not** independent. Assuming independence dramatically underestimates simultaneous failure risk.

# The Independence Test, Drawn



# Part 4

## Bayes' Theorem

# The Problem: You Can Only Go One Way

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With COUNTIFS, you can compute  $P(A \mid B) = \text{Count}(A \text{ AND } B) / \text{Count}(B)$ .

But some engineering problems give you  $P(A \mid B)$  and ask for  $P(B \mid A)$ . These are not the same, and you cannot get one from the other without more information.

## HVAC fault detection scenario:

You know how good the fault detection algorithm is:

- $P(\text{Alarm} \mid \text{Fault}) = \text{true positive rate}$  — alarm fires when the fault is real (correct detection)
- $P(\text{Alarm} \mid \text{No Fault}) = \text{false positive rate}$  — alarm fires when no fault exists (false alarm)

But what the maintenance team needs is the **reverse**:

- **$P(\text{Fault} \mid \text{Alarm})$**  — given that the alarm fired, is there really a fault?

# The Same Table, Read Two Ways

Same 1,000 hours, same 45 — a different denominator

Across the row — what you are given

|          | Alarm | No alarm |
|----------|-------|----------|
| Fault    | 45    | 5        |
| No fault | 143   | 807      |

$P(\text{Alarm} \mid \text{Fault}) = 45 / 50 = 90\%$

Down the column — what you need

|          | Alarm | No alarm |
|----------|-------|----------|
| Fault    | 45    | 5        |
| No fault | 143   | 807      |

$P(\text{Fault} \mid \text{Alarm}) = 45 / 188 = 24\%$

# Bayes' Theorem

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$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A)}$$

*Read as: "updated probability of B = (how likely the evidence is when B is true) × (prior chance of B), divided by the overall chance of seeing the evidence."*

## The three ingredients:

| Term     | Name              | Meaning                                                          |
|----------|-------------------|------------------------------------------------------------------|
| P(B)     | <b>Prior</b>      | How common is the condition before any evidence                  |
| P(A   B) | <b>Likelihood</b> | How likely is the observation if the condition is present        |
| P(B   A) | <b>Posterior</b>  | Revised probability of the condition given the observed evidence |



## Bayes' Theorem — The Denominator

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The denominator  $P(A)$  in Bayes' theorem expands using the **total probability rule**:

$$P(A) = P(A \mid B) \cdot P(B) + P(A \mid B^c) \cdot P(B^c)$$

*Read as: "overall chance of evidence  $A$  = (chance of  $A$  when  $B$  is true  $\times$  how common  $B$  is) + (chance of  $A$  when  $B$  is false  $\times$  how common 'not  $B$ ' is)."*

# HVAC Fault Detection — Setting Up the Problem

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A building automation system monitors a Tucson office building's chiller. Based on maintenance records and algorithm documentation:

| What we know                                                               | Value       |
|----------------------------------------------------------------------------|-------------|
| $P(\text{Fault})$ — faults occur in 5% of summer operating hours           | <b>0.05</b> |
| $P(\text{Alarm} \mid \text{Fault})$ — algorithm catches 90% of real faults | <b>0.90</b> |
| $P(\text{Alarm} \mid \text{No Fault})$ — 15% false positive rate           | <b>0.15</b> |

**What the maintenance team needs:**  $P(\text{Fault} \mid \text{Alarm}) = ?$

*Before computing — what is your intuition? The algorithm is 90% accurate. Does that mean 90% of alarms are real faults?*

# Bayes' Theorem — Computing $P(\text{Fault} \mid \text{Alarm})$

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**Step 1 — Total probability of an alarm:**

$$\begin{aligned} P(\text{Alarm}) &= P(\text{Alarm} \mid \text{Fault}) \cdot P(\text{Fault}) + P(\text{Alarm} \mid \text{No Fault}) \cdot P(\text{No Fault}) \\ &= 0.90 \times 0.05 + 0.15 \times 0.95 = 0.045 + 0.1425 = \mathbf{0.1875} \end{aligned}$$

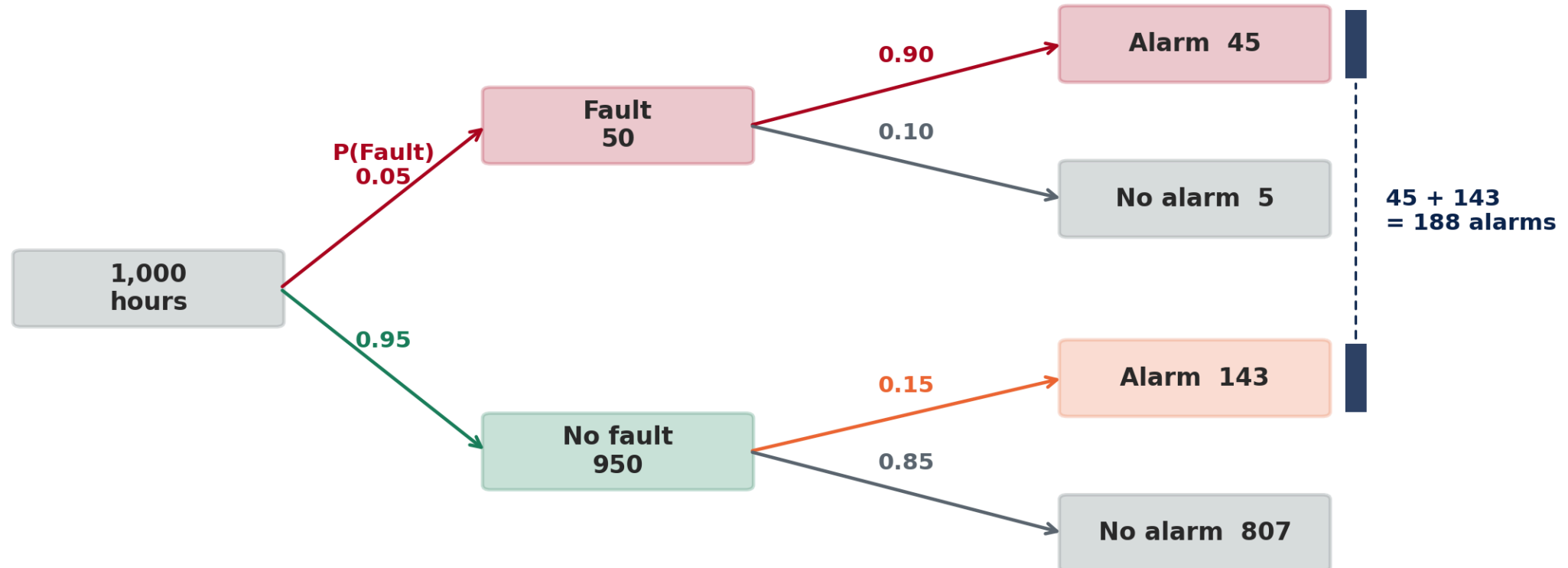
**Step 2 — Apply Bayes:**

$$P(\text{Fault} \mid \text{Alarm}) = \frac{P(\text{Alarm} \mid \text{Fault}) \cdot P(\text{Fault})}{P(\text{Alarm})} = \frac{0.045}{0.1875} = \mathbf{0.24}$$

Only **24% of alarms** correspond to real faults — even though the algorithm detects 90% of faults when they exist.

# The Same Calculation as a Tree

Only the two 'Alarm' leaves are still in play



$$P(\text{Fault} \mid \text{Alarm}) = 45 / (45 + 143) = 24\%$$

## Verify With a Frequency Table

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*Here is the same calculation using concrete counts — imagining 1,000 summer operating hours makes the fractions tangible.*

**Imagine 1,000 summer operating hours:**

|               | Alarm fires | No alarm | Row total        |
|---------------|-------------|----------|------------------|
| Fault present | 45          | 5        | 50 (5% of 1,000) |
| No fault      | 143         | 807      | 950              |
| Column total  | 188         | 812      | 1,000            |

$$P(\text{Fault} \mid \text{Alarm}) = \frac{45}{188} = 23.9\% \approx 24\% \checkmark$$

The algorithm fires **188 alarms per 1,000 hours**. Only **45 are real** — **76% of alarms are noise**.

# Why the Result Surprises Everyone — The Base Rate

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The 90% detection rate sounds impressive. Why is only 24% of alarms real?

**Because faults are rare (5% base rate), so false positives dominate:**

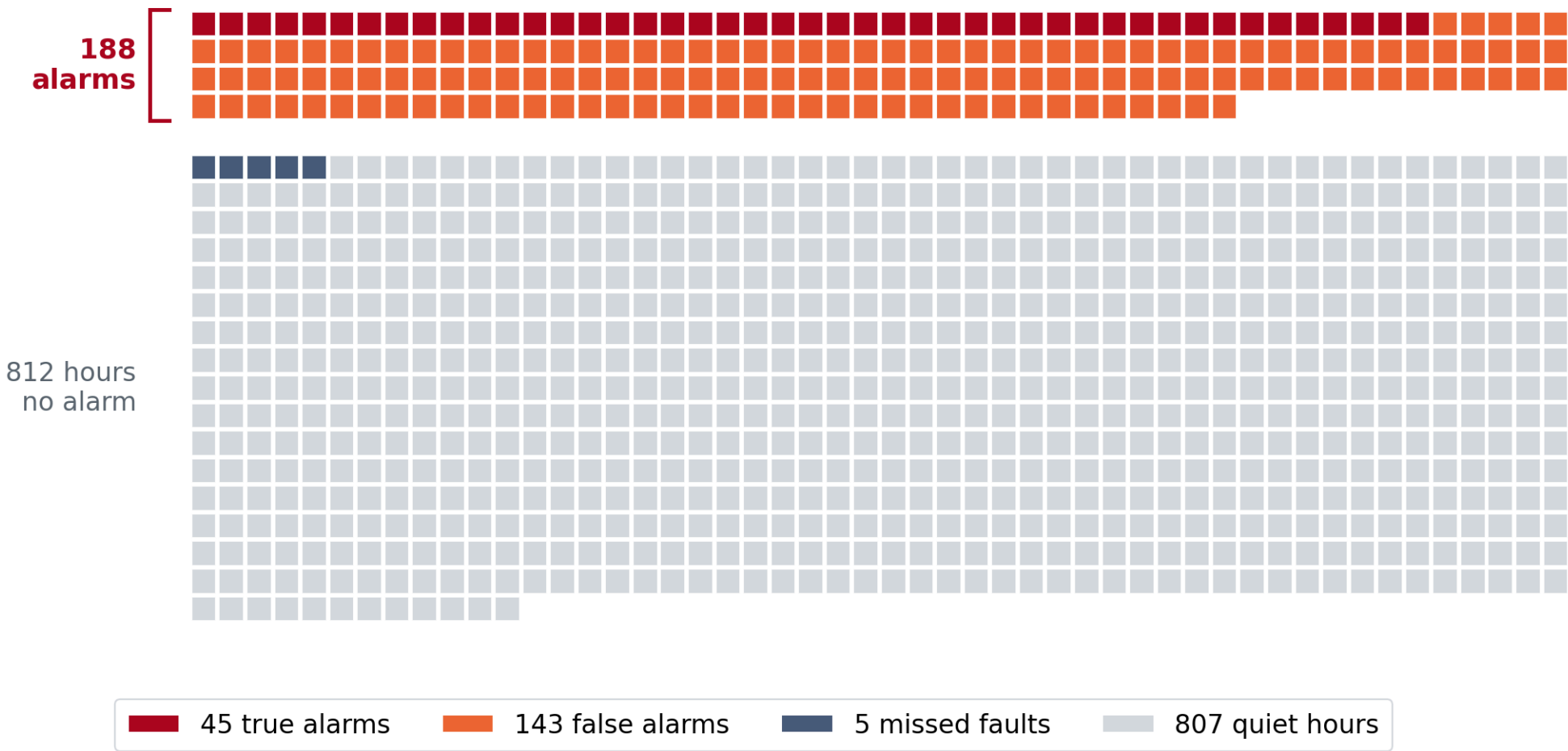
- 950 fault-free hours  $\times$  15% false positive rate = **143 false alarms**
- 50 faulty hours  $\times$  90% detection = **45 true alarms**

The false alarms outnumber the real ones 3-to-1 simply because there are 19 $\times$  more fault-free hours than faulty ones.

 **The base rate effect:** Even a highly accurate test produces mostly false positives when the condition it tests for is rare. This is why medical screening tests, structural inspection protocols, and building fault detection systems all report precision (what fraction of positives are true) alongside sensitivity (what fraction of true positives are detected). Bayes' theorem is the bridge between the two.

# One Thousand Hours, One Square Each

1,000 summer hours · a 90%-accurate detector · 76% of its alarms are noise



# Engineering Response — What Does 24% Mean?

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$P(\text{Fault} \mid \text{Alarm}) = 24\%$  is not negligible — and not actionable as a confirmed fault.

**The tiered-response protocol that Bayes justifies:**

| Posterior probability                      | Recommended response                           |
|--------------------------------------------|------------------------------------------------|
| $P(\text{Fault} \mid \text{Alarm}) < 10\%$ | Log, do not dispatch                           |
| 10–40%                                     | Send technician to inspect (not repair)        |
| 40–70%                                     | Dispatch with parts; inspect before installing |
| $> 70\%$                                   | Treat as confirmed fault; repair on arrival    |

At 24%, the right call is **inspection, not emergency repair**. This is how modern CMMS (computerized maintenance management systems) use Bayesian fault scoring to prioritize work orders — not by binary alarm/no-alarm, but by posterior probability.



## Conditional Probability — Common Mistakes

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⚠  $P(A | B)$  and  $P(B | A)$  use the same joint count in the numerator — they differ only in the denominator. Swapping numerator and denominator is the single most common error when working with conditional probabilities.

## Quick Check — Apply Bayes Yourself

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A structural health monitoring sensor is installed on a bridge deck joint. Based on inspection records and sensor specifications:

| What we know                                                              | Value       |
|---------------------------------------------------------------------------|-------------|
| $P(\text{Defect})$ — 2% of monitored joints have active cracks            | <b>0.02</b> |
| $P(\text{Alert} \mid \text{Defect})$ — sensor detects 80% of real defects | <b>0.80</b> |
| $P(\text{Alert} \mid \text{No Defect})$ — 5% false positive rate          | <b>0.05</b> |

**Compute:**  $P(\text{Defect} \mid \text{Alert}) = ?$

*Apply the two-step Bayes calculation. Work individually for 3 minutes, then compare with your neighbor.*

## Quick Check — Answer

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**Step 1 — Total probability of an alert:**

$$P(\text{Alert}) = 0.80 \times 0.02 + 0.05 \times 0.98 = 0.016 + 0.049 = \mathbf{0.065}$$

**Step 2 — Bayes:**

$$P(\text{Defect} \mid \text{Alert}) = \frac{0.016}{0.065} \approx \mathbf{24.6\%}$$

Only **1 in 4 alerts** corresponds to a real defect — even with an 80% detection rate.

**Same result as the HVAC example (24%)** — when the condition is rare, false positives from the majority dominate regardless of domain.

# Part 5

Today's Lab

# What You'll Do Today — Section A

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**Dataset:** `Week3_Tucson_Office_Energy_2023.csv` — 8,760 hourly rows, 12 columns

## Section A — Marginal Probabilities

- A1–A2: Count hours with any cooling load and with high cooling demand ( $> 50$  kWh)
- A3: Marginal  $P(\text{Cooling} > 50)$  — your baseline for all comparisons
- A4–A5: High total electricity counts and marginal  $P(\text{Electricity} > 100)$
- A6–A7: Count summer hours and weekday hours
- A8: Average July electricity — first hint of what conditioning reveals

# What You'll Do Today — Section B

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## Section B — Conditional Probabilities and Independence

- B1–B2:  $P(\text{Cooling} > 50 \mid \text{July})$  — your first full conditional probability
- B3:  $P(\text{Cooling} > 50 \mid \text{January})$  — the winter baseline
- B4: Independence test for Cooling and Month
- B5–B6:  $P(\text{Electricity} > 100 \mid \text{Weekday})$
- B7–B10: Weekend and business-hours conditioning
- B11–B13: Pivot table —  $P(\text{Cooling} > 50 \mid \text{Season})$ , peak month, mean under peak conditions
- B14: Reverse conditioning —  $P(\text{Weekday} \mid \text{July})$  — sets up Bayes' theorem

## In-Lab Target — What to Finish Before Class Ends

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### Take-home (due before next class):

- B11–B14: Pivot table analysis, peak month, mean under peak conditions, reverse conditioning
- Chart 2: Conditioning effect comparison (5 bars)
- Memo PDF: 3 paragraphs on marginal vs. conditional probability for HVAC sizing
- 💡 **B14:**  $P(\text{Weekday} \mid \text{July})$  flips the conditioning — Bayes' theorem formalizes this.

# Looking Ahead — Week 4 Preview

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At the end of today's lab you will see this question:

*"If extreme-cooling hours occur at an average rate of  $\lambda = 4$  per week during July, what is the probability a randomly chosen July week has exactly 6 extreme-cooling hours?"*

## The question to sit with:

- Is cooling demand exactly 4 per week every week, or does it vary?
- What statistical distribution models the **count of rare events** that occur at a known average rate?

💡 Week 4 is the **Poisson distribution** — the natural model for counts of events (equipment faults, monsoon rain days, building occupancy spikes) that occur at a roughly constant average rate. It builds directly on the probabilities you compute today.



# Summary — Week 3

**Conditioning is not a technicality — it is the difference between the right answer and a dangerous approximation**

**Today's concepts:**

- **Classical probability** — complement, addition rule, independence
- **Conditional probability** —  $P(A \mid B) = \text{joint count} / \text{group count}$ ; restricts the sample space
- **Independence test** — compare joint probability to product of marginals
- **Bayes' theorem** — reverse the conditional; update a prior with new evidence

**The core insight:** Marginal probabilities are weighted averages. Conditioning on context (season, occupancy, fault state) reveals the structure that design decisions require.

# Key Formulas – Probability Rules

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| Concept                 | Formula                                            |
|-------------------------|----------------------------------------------------|
| Complement              | $P(A^c) = 1 - P(A)$                                |
| Addition rule           | $P(A \cup B) = P(A) + P(B) - P(A \cap B)$          |
| Conditional probability | $P(A   B) = P(A \cap B) / P(B)$                    |
| Independence test       | $P(A \cap B) = P(A) \times P(B)$ if independent    |
| Total probability       | $P(A) = P(A B) \cdot P(B) + P(A B^c) \cdot P(B^c)$ |

## Key Formulas — Bayes' Theorem & Reference

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| Concept        | Formula                             |
|----------------|-------------------------------------|
| Bayes' theorem | $P(B A) = P(A B) \cdot P(B) / P(A)$ |